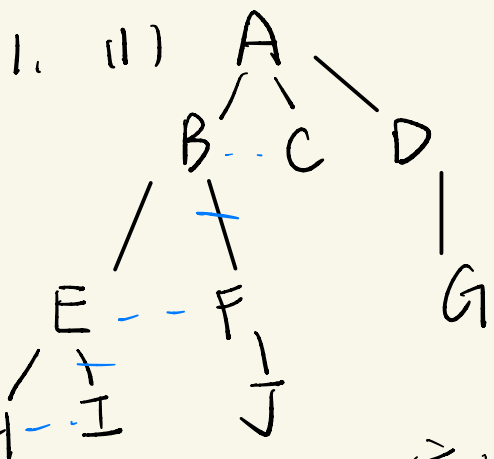


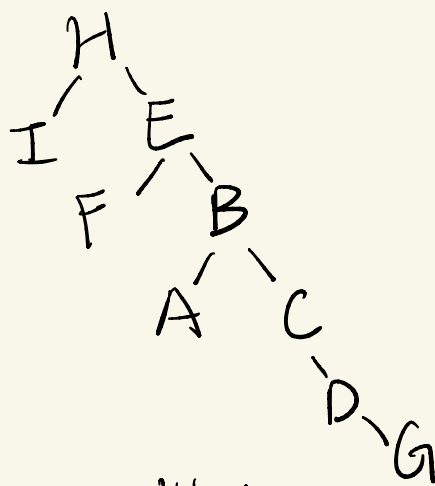
六



$(A(B(E(H,I), F(J)), C(D(G)))$

(2) 深3 高4

(3)



```

if kNode.child is not empty:
    Node.left = convert(Node.child)
    pos = Node.right
    for i from 1 to length:
        right = convert(Node)
        pos = pos.right
    
```

2. (1)  $\left[ \frac{k^k - 1}{k - 1}, \frac{k^{k+1} - 1}{k - 1} \right]$

(2)  $\left\lfloor \frac{N-1}{k} \right\rfloor$

(3)  $k \cdot N + i + 1$

3. 证明: 当  $n=1$  时, 深度为0, 满足

假设  $n=k$  时成立

$n=k+1$  时 合并两集合时, 假设高度分别为  $h_1, h_2$

且  $h_1 \leq h_2$   $\begin{cases} \text{若 } h_1 < h_2 & \text{合并后树高仍为 } h_2 \\ h_1 = h_2 & \text{合并后树高 } h_2 + 1 \end{cases}$

则每次 union

树高最多+1, 则进行  $k+1$  次

树高最大为  $\log_2(k+1)$  证